WORKED SOLUTION

STEP I 1996, Question 7

Pure • First-Order Differential Equations

STEP I 1996, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q7

The key idea

"Flows out at a rate proportional to the amount" translates directly into a first-order differential equation, $\frac{\mathrm{d}y}{\mathrm{d}t} = -ky$ with $k > 0$, whose solution is an exponential. Writing e^{-k} as a single number b turns the exponential into b^t, and the same substitution $k = -\ln b$ rewrites the equation in part (ii). That equation is linear, so an integrating factor finishes it.

Solution

Part (i)

The water flows *out*, so y decreases at a rate proportional to y:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -ky \qquad \text{for some constant } k > 0.$$

While there is water in the tank, $y > 0$, so we may separate the variables:

$$\int \frac{\mathrm{d}y}{y} = -\int k\,\mathrm{d}t \quad \Longrightarrow \quad \ln y = -kt + c.$$

At $t = 0$ we have $y = 1$, so $c = \ln 1 = 0$. Hence

$$y = \mathrm{e}^{-kt} = \left(\mathrm{e}^{-k}\right)^t = b^t, \qquad \text{where } b = \mathrm{e}^{-k}.$$

Since $k > 0$, we have $0 < b < 1$; in particular $b < 1$, as required. Note for later that

$$\ln b = -k, \qquad \text{so} \qquad k = -\ln b > 0.$$

Part (ii)

The outflow is exactly as before, at rate ky with the same constant k, and in addition water comes in at rate a. So the net rate of change is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = a - ky.$$

? **Ask yourself** Where does $\ln b$ come from, when this equation only involves k?

The constant b was *defined* in part (i) by $b = \mathrm{e}^{-k}$, so it describes the outflow and nothing else; it is still the same number here. Substituting $k = -\ln b$ gives $\frac{\mathrm{d}y}{\mathrm{d}t} = a + y\ln b$, that is,

$$\frac{\mathrm{d}y}{\mathrm{d}t} - y\ln b = a,$$

which is the required equation.

This is linear in y with constant coefficients. The integrating factor is

$$\exp\left(-\int \ln b \,\mathrm{d}t\right) = \mathrm{e}^{-t\ln b} = b^{-t}.$$

Multiplying through by b^{-t}, the left-hand side becomes an exact derivative, because $\frac{\mathrm{d}}{\mathrm{d}t}b^{-t} = -b^{-t}\ln b$:

$$\frac{\mathrm{d}}{\mathrm{d}t}(y\,b^{-t}) = a\,b^{-t}.$$

Integrating, and using $\int b^{-t}\,\mathrm{d}t = -\dfrac{b^{-t}}{\ln b}$ (valid since $\ln b \neq 0$),

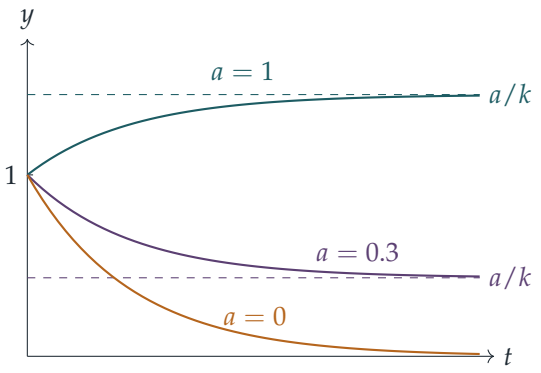
$$y\,b^{-t} = -\frac{a}{\ln b}\,b^{-t} + C.$$

At $t = 0$, $y = 1$, so $1 = -\dfrac{a}{\ln b} + C$, giving $C = 1 + \dfrac{a}{\ln b}$. Multiplying through by b^t,

$$\begin{aligned} y &= -\frac{a}{\ln b} + \left(1 + \frac{a}{\ln b}\right) b^t \\ &= b^t + \frac{a\,(1 - b^t)}{-\ln b}. \end{aligned}$$

Both forms are fine. The second is easier to read, because $-\ln b = \ln(1/b) > 0$: it is the part (i) solution b^t plus a positive term coming from the inflow.

As $t \to \infty$, $b^t \to 0$ (since $0 < b < 1$), so $y \to \dfrac{a}{-\ln b} = \dfrac{a}{k}$. This is the equilibrium level, at which the outflow ky exactly balances the inflow a. The tank fills towards this level if $a > k$ and drains towards it if $a < k$.



The sketch shows $b = \frac{1}{2}$ (so $k = \ln 2 \approx 0.693$) with $a = 1$, $a = 0.3$ and, for comparison, the part (i) curve $a = 0$.

Checks

- Put $a = 0$: the answer reduces to $y = b^t$, the part (i) result.
- At $t = 0$: $y = 1 + \frac{a(1-1)}{-\ln b} = 1$.
- Differentiate $y = b^t + \frac{a}{\ln b}(b^t - 1)$: we get $y' = b^t \ln b + ab^t$, while $y \ln b + a = b^t \ln b + a(b^t - 1) + a = b^t \ln b + ab^t$, which agrees.
- Numerically, with $b = \frac{1}{2}$, $a = 1$, $t = 2$, the formula gives $y \approx 1.3320$; stepping the differential equation with a Runge–Kutta method gives the same value, and in general agreement to about 10^{-13} for random a, b, t.

Common slips

- Writing $\frac{\mathrm{d}y}{\mathrm{d}t} = ky$ and then forgetting that k must be negative for an outflow; the sign is what forces $b < 1$.
- Treating b as a new unknown in part (ii). It is fixed by the outflow alone, so $\ln b = -k$ carries straight over.
- Getting the sign wrong in $\int b^{-t}\,\mathrm{d}t = -b^{-t} / \ln b$; check by differentiating.

Beyond STEP

This is the simplest compartment model: one tank with a constant input and a proportional output. The same equation describes a drug infused into the bloodstream at a steady rate and cleared by the kidneys, or a capacitor charging through a resistor. Every solution relaxes exponentially to the equilibrium a/k, which is an example of a stable fixed point, an idea that becomes central when you meet systems of differential

equations. Boyce and DiPrima, *Elementary Differential Equations*, treats linear first-order equations and many such mixing problems.

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